Table 1. The 134 core targets (scheme A) and their layered evidence.

Gene	SEA R0 metabolites	GeneCards relevance	In DisGeNET	STRING degree	Hub (5-alg. consensus)	KEGG terms (BH<0.05)	NPC log2FC	NPC meta FDR	Robust DEG	Malignant fold (scRNA)	gutMGene records (supported/total)	Docked
ABCB1	4	227.2	no	32		4	-0.71	1.2e-04	no	0.09	—	
ABCB11	2	89.1	no	14		3	-0.19	0.017	no	0.00	—	
ABCC1	6	189.6	no	14		2	n/a	n/a	n/a	3.09	—	
ABCG2	7	113.4	yes	32		2	-0.41	0.014	no	0.04	—	
ABL1	1	218.6	no	27		9	+0.12	0.428	no	1.40	—	
ACE2	1	125.9	no	14		1	+0.37	0.393	no	4.06	0/3	
ADAMTS4	1	31.4	no	7		0	+0.44	0.100	no	0.71	—	
ADRA2B	3	39.6	no	2		0	+0.23	0.774	no	1.34	1/1	
AHR	1	83.0	no	5		3	-0.24	0.350	no	1.38	5/5	
AKR1B1	5	58.0	no	15		2	+0.38	0.022	no	1.07	—	
AKR1B10	2	n/a	yes	7		2	-0.01	0.558	no	6.01	—	
AKR1C3	1	74.9	no	10		6	-2.04	6.5e-09	yes	0.45	—	
AKT1	1	436.6	yes	74	hub	80	-0.23	0.018	no	0.82	3/3	
ALDH2	1	141.6	no	10		3	+0.10	0.653	no	2.05	—	
ALK	1	301.2	no	14		3	+0.60	0.010	no	1.14	—	
ALOX12	2	49.7	no	8		4	+0.40	0.612	no	8.29	—	
ALOX15	2	64.1	no	12		5	-1.32	0.069	no	0.35	—	
ALOX5	5	86.1	no	15		6	-0.13	0.320	no	0.17	—	
ALPL	2	93.0	no	5		2	-0.32	0.017	no	0.14	—	
APEX1	1	122.6	no	10		0	+0.12	0.118	no	1.50	—	
AR	2	270.0	no	31		4	-0.64	4.1e-04	no	2.22	—	
BACE1	4	53.7	no	13		1	-0.18	0.075	no	0.94	—	
BRAF	2	454.1	no	27		38	-0.01	0.414	no	1.34	—	
CA5B	3	13.0	no	0		1	-0.77	4.1e-06	no	0.53	—	
CA9	7	85.6	no	7		1	+1.20	4.5e-04	yes	11.81	—	yes
CACNA1H	1	58.1	no	3		4	+0.05	0.071	no	1.38	—	
CAMK2A	1	39.2	no	7		20	-0.25	0.119	no	8.34	1/1	
CASP7	1	77.9	no	10		8	-0.17	0.350	no	0.98	—	
CBR1	2	61.0	no	13		6	-0.45	0.042	no	2.77	—	
CD38	1	101.2	no	8		3	-0.02	0.901	no	0.09	—	
CDK1	1	105.2	no	26		4	+1.31	3.4e-08	yes	2.47	—	
CDK6	2	214.7	no	18		18	+0.16	0.284	no	0.31	1/1	
CDK9	1	90.3	no	6		1	-0.14	0.013	no	0.90	—	
CFTR	2	238.2	no	12		4	-0.67	0.003	no	3.44	—	
CHRM2	2	25.1	no	4		5	-0.14	0.877	no	n/a	1/1	
CHRNA7	1	69.3	no	0		4	n/a	n/a	n/a	3.31	—	
CISD1	2	15.8	no	3		1	-0.07	0.801	no	2.16	—	
CNR1	3	53.7	no	8		1	-0.70	1.7e-04	no	0.27	1/1	
CNR2	1	32.1	no	4		0	-0.53	0.059	no	0.04	—	
CSNK2A1	2	72.8	no	12		4	+0.31	8.8e-05	no	1.81	—	
CTNNB1	1	409.7	yes	65	hub	27	-0.02	0.946	no	1.14	3/3	
CYP1A1	8	119.9	yes	23		10	-0.13	0.325	no	13.73	5/6	
CYP1A2	12	104.8	no	20		10	+0.21	0.034	no	25.11	3/3	
CYP1B1	9	112.5	no	13		8	-0.59	0.020	no	1.48	—	
CYP2A6	1	114.4	no	15		6	-0.95	9.4e-05	no	0.74	—	
CYP2C19	9	107.3	no	22		6	+0.01	0.847	no	1.12	—	
CYP2C8	1	112.9	no	17		8	-0.77	1.1e-04	no	1.19	—	
CYP2C9	6	138.6	no	24		9	-0.17	0.921	no	5.68	—	
CYP2D6	1	152.4	no	17		4	-0.08	0.289	no	1.44	1/1	
CYP3A4	10	139.8	no	29		9	+0.46	0.104	no	3.61	1/1	
DAPK1	2	71.3	no	1		2	+0.04	0.881	no	0.65	—	
DPP4	7	102.6	no	17		0	-0.15	0.595	no	0.08	—	
EGFR	3	434.3	yes	56	hub	44	+0.52	0.007	no	10.59	1/4	
ELAVL1	1	58.9	no	17		2	+0.12	0.010	no	1.37	—	

Gene	SEA R0 metabolites	GeneCards relevance	In DisGeNET	STRING degree	Hub (5-alg. consensus)	KEGG terms (BH<0.05)	NPC log2FC	NPC meta FDR	Robust DEG	Malignant fold (scRNA)	gutMGene records (supported/total)	Docked
ENPP2	1	52.1	no	1		1	-0.34	0.282	no	0.60	—	
ESRRA	2	69.5	no	8		0	+0.07	0.589	no	2.00	—	
F2	1	98.2	no	4		5	+0.31	0.437	no	0.81	—	
FAAH	1	25.4	no	9		0	-0.41	0.004	no	3.86	—	
GLO1	3	62.9	no	7		1	+0.34	0.005	no	2.43	—	
GPBAR1	7	45.1	no	8		0	+0.12	0.996	no	0.14	1/2	
GSK3B	3	180.9	no	48	hub	38	+0.19	0.047	no	1.25	1/1	
GSTK1	1	24.3	no	2		4	-0.36	3.6e-04	no	0.66	—	
HDAC3	1	67.5	no	16		3	+0.04	0.892	no	0.84	3/4	
HDAC6	10	102.3	no	25		2	-0.15	0.034	no	0.89	1/1	
HRH1	2	32.5	no	2		2	-0.38	0.022	no	1.06	—	
HTR2B	1	31.1	no	5		3	+0.22	0.564	no	6.22	—	
IDE	1	32.3	no	10		1	+0.11	0.097	no	2.00	—	
IGF1R	1	213.0	yes	24		25	+0.20	0.179	no	6.30	0/2	
IP6K2	7	21.9	no	0		1	-0.18	0.346	no	0.98	—	
IPMK	5	48.3	no	0		1	-0.24	0.300	no	1.21	—	
KCNA5	1	53.1	no	4		0	-0.39	0.058	no	0.01	—	
KCND3	1	40.6	no	3		0	-0.20	0.249	no	6.05	—	
KDM1A	2	132.0	yes	21		0	+0.14	0.122	no	2.14	—	
KDM4E	2	18.1	no	2		0	n/a	n/a	n/a	n/a	—	
KDM5B	1	53.9	no	6		0	+0.44	0.002	no	2.47	—	
KDM6B	1	63.2	no	12		0	+0.14	0.580	no	0.45	—	
KDR	2	267.9	no	29		12	-0.26	0.719	no	0.90	—	
MAOA	8	63.5	no	20		4	-0.66	0.019	no	7.55	—	
MAOB	3	31.8	no	20		4	-1.54	9.4e-10	yes	0.61	—	yes
MAP2K1	2	246.8	yes	29		73	+0.15	0.058	no	0.85	—	
MAP2K4	1	130.4	yes	9		17	+0.08	0.482	no	1.43	—	
MAPT	1	125.5	no	18		2	-0.39	0.356	no	6.55	—	
MBNL1	1	35.2	no	2		0	-0.12	0.036	no	0.60	—	
MCL1	1	158.2	no	33		3	-0.10	0.524	no	0.76	—	
MET	1	369.8	no	26		22	-0.52	0.176	no	5.69	—	
MKNK1	1	81.6	no	3		3	+0.10	0.357	no	1.00	—	
MKNK2	2	65.7	no	2		3	-0.05	0.164	no	1.54	—	
MMP1	1	117.6	yes	15		7	+3.27	5.9e-12	yes	1.62	—	
MMP13	1	100.5	no	13		3	+0.94	0.005	no	1.06	—	
MMP2	3	170.4	yes	30		10	+0.45	0.005	no	0.09	—	
MMP3	1	63.6	no	14		6	+3.28	8.4e-11	yes	0.51	1/1	
MMP9	3	197.6	yes	41		16	+0.19	0.498	no	0.29	6/6	
NEK2	1	131.3	no	7		0	+0.89	1.4e-05	no	2.89	—	
NEK6	1	60.3	no	0		0	+0.23	0.425	no	0.91	—	
NLRP3	1	153.8	no	12		6	-0.19	0.254	no	0.07	5/8	
P4HB	2	50.3	no	4		0	+0.07	0.779	no	1.40	—	
PARP1	1	221.9	yes	37		2	+0.23	0.338	no	1.26	—	
PDE4D	1	63.2	no	4		3	+0.03	0.626	no	0.76	0/1	
PDE5A	1	45.9	no	2		1	-0.19	0.340	no	1.73	—	
PIK3CA	1	457.8	yes	28		84	+0.01	0.898	no	1.25	—	
PIK3CG	1	152.4	no	19		9	-0.67	0.005	no	0.07	—	
PIK3R1	1	203.5	no	29		82	-0.35	0.003	no	0.56	—	
PIM1	2	205.9	no	2		3	-0.15	0.131	no	0.29	—	
PKM	5	68.4	no	20		5	+0.43	2.1e-04	no	1.30	—	
PKN1	1	68.3	no	3		3	-0.36	9.6e-04	no	0.30	—	
PLAA	2	56.0	no	2		0	+0.28	0.001	no	1.21	—	
PLK1	1	97.5	yes	10		3	+0.29	0.429	no	1.46	—	
PPARA	3	108.8	no	32		9	-0.39	0.012	no	1.56	2/3	
PPARD	2	93.3	no	14		4	-0.15	0.018	no	0.96	—	
PPARG	3	219.9	no	52		8	-0.20	0.103	no	0.26	4/4	
PSMB5	6	50.3	no	5		1	+0.21	0.022	no	2.51	—	

Gene	SEA R0 metabolites	GeneCards relevance	In DisGeNET	STRING degree	Hub (5-alg. consensus)	KEGG terms (BH<0.05)	NPC log2FC	NPC meta FDR	Robust DEG	Malignant fold (scRNA)	gutMGene records (supported/total)	Docked
PTGS1	7	128.3	no	18		5	+0.07	0.889	no	0.15	—	
PTGS2	2	190.5	yes	53	hub	19	+2.45	1.9e-10	yes	6.18	4/5	yes
PTK2	1	160.2	no	20		26	+0.40	0.002	no	3.14	—	
PTPN1	3	106.4	no	17		4	-0.07	0.516	no	0.60	—	
PYGL	1	53.2	no	5		4	-0.37	0.019	no	0.60	—	
RPS6KA3	2	105.3	no	8		8	-0.25	6.9e-04	no	0.60	—	
RXRA	1	115.1	no	14		16	-0.09	0.841	no	1.72	—	
SETD7	1	65.9	no	4		2	-0.18	0.131	no	1.71	—	
SIRT6	2	73.2	no	12		2	+0.09	0.204	no	1.46	—	
SLC1A1	1	39.9	no	1		0	-0.87	7.4e-05	no	2.96	—	
SLC2A1	2	136.2	no	22		10	+0.49	0.034	no	2.19	1/1	
SLCO2B1	3	60.6	no	10		0	-0.00	0.974	no	0.64	—	
SNCA	2	119.9	no	19		1	-0.25	0.295	no	1.74	—	
SPHK1	1	47.2	no	6		7	+0.64	0.001	no	0.75	—	
SRC	2	250.6	yes	56	hub	36	+0.42	0.003	no	1.68	—	
SYK	3	193.4	yes	24		12	-0.48	0.010	no	0.55	—	
TERT	1	455.9	yes	21		4	-0.00	0.881	no	2.12	—	
TLR2	1	156.2	yes	20		13	+0.35	0.204	no	0.43	6/6	
TLR4	1	193.3	no	33		17	+0.00	0.536	no	0.05	5/7	
TNKS	2	50.3	no	7		0	+0.19	0.252	no	1.74	—	
TOP2A	1	169.3	no	14		0	+1.47	3.1e-06	yes	2.57	—	
VDR	1	189.0	no	0		3	+0.62	3.2e-04	no	0.61	1/1	yes
YES1	3	110.6	no	12		1	+0.54	3.5e-04	no	4.01	—	

Note. Core targets = SEA R0 (exact known ChEMBL36 ligand, Homo; records whose ChEMBL structure differs from the metabolite by InChIKey removed, scheme A) ∩ STP ∩ disease genes. STRING degree: STRING v12.0, score ≥ 0.4, 128 connected targets (six isolated targets have degree 0). Hub: top-10 intersection of five centrality algorithms (cytoHubba definitions re-implemented; EPC not recomputed). NPC log2FC / FDR: three-array meta-analysis (n/a = not measurable); robust DEG: FDR < 0.05, |log2FC| ≥ 1 and the same direction in 3/3 arrays. Malignant fold: malignant-cell expression relative to other cell types (single-cell dataset, E49). gutMGene records: every gutMGene record for that target was checked against the source paper; the cell gives records supported by the source text / records checked (E90, 87 records for 30 targets). "—" = the target has no gutMGene record. Systems used by the source (animal in vivo vs in vitro) are reported per record in TableS_gutMGene; this column is a count, not a ranking of evidence strength. Counts are records, not independent studies: several records may come from the same paper. Network-central genes are common across many diseases.

Table 2. The six hub genes and their ranks in five centrality algorithms (scheme A network, 128 nodes / 1,037 edges).

Gene	Rank MCC	Rank MNC	Rank EPC	Rank Stress	Rank Degree	Rank Radiality	STRING degree	gutMGene records (supported/total)	NPC log2FC	NPC meta FDR
AKT1	2	1	n/a	1	1	1	74	3/3	-0.23	0.018
CTNNB1	1	2	n/a	2	2	2	65	3/3	-0.02	0.946
EGFR	3	3	n/a	5	3	3	56	1/4	+0.52	0.007
GSK3B	9	7	n/a	7	7	7	48	1/1	+0.19	0.047
PTGS2	7	5	n/a	3	5	5	53	4/5	+2.45	1.9e-10
SRC	4	4	n/a	4	4	4	56	—	+0.42	0.003

Note. All six genes appear in all five top-10 lists. Ranks recomputed in networkx with cytoHubba definitions; EPC (stochastic, ties uninformative; cytoHubba not scriptable) not recomputed, n/a. Hubs were selected by network centrality, not by disease specificity; only PTGS2 passes the robust differential-expression threshold in NPC. gutMGene records as in Table 1.

Table 3. Nine metabolite–target docking pairs (AutoDock Vina) with redocking validation and blood-concentration source.

#	Metabolite	Target	PDB	Co-crystal ligand	Co-crystal redock	Co-crystal RMSD (Å)	Vina top pose (kcal/mol)	Pose energy range	Contact residues (4 Å)	SEA record (scheme A)	Geometric H-bonds	Blood upper bound (μM)	R (total conc.)	Conc. source
1	Quercetin	PTGS2	5IKQ	JMS	pass	0.58	-7.76	-7.76 ~ -6.45	16	correct	8	0.255	0.238	OK
2	Apigenin	PTGS2	5IKQ	JMS	pass	0.58	-7.75	-7.75 ~ -6.72	13	correct	3	0.208	0.026	DOSE
3	Apigenin	MAOB	2V5Z	SAG	pass	1.39	-10.00	-10.00 ~ -8.45	11	correct	3	0.208	0.185	DOSE
4	Genistein	MAOB	2V5Z	SAG	pass	1.39	-9.65	-9.65 ~ -8.61	11	correct	1	0.985	1.53	OK
5	Acacetin	MAOB	2V5Z	SAG	pass	1.39	-10.23	-10.23 ~ -8.12	14	correct	3	0.0018	0.037	MANUAL
6	Quercetin	CA9	6FE0	V90	pass	0.90	-7.98	-7.98 ~ -6.54	17	correct	9	0.255	0.037	OK
8	Dopamine	ADRA2A	9CBL	ALE	fail	2.01	-6.10	-6.10 ~ -5.48	11	correct	7	0.00013	0	UNK
9	Levodopa	ADRA2A	9CBL	ALE	fail	2.01	-6.92	-6.92 ~ -6.24	13	correct	8	n/a	n/a	—
10	Lithocholic Acid	VDR	3AZ3	DS6	pass	1.18	-10.72	-10.72 ~ -7.95	21	correct	0	0.37	1.12	PED

Note. Co-crystal redock check: the receptor's co-crystal ligand is re-docked into its own structure; pass = RMSD ≤ 2.0 Å. This validates the docking protocol, not the query metabolite's pose. For VDR (3AZ3) the co-crystal ligand is DS6, not lithocholic acid; the LCA pose is a cross-dock and is not validated by this check. The two ADRA2A pairs failed the redock check (shown in FigS_docking_failed_redock) and are hypothesis-generating only. The decanoic acid–GPR84 pair is excluded: its SEA record is a different structure (InChIKey check), so the edge does not exist in scheme A; it is reported in TableS_docking_excluded_pair and marked in Figs 5 and 5b. Row numbers follow the original docking run, so no. 7 is the excluded pair. Vina energies are scores, not measured affinities. Blood upper bound and R (blood upper bound / strongest known activity) are from the concentration filter on total concentration; conc. source: OK = adult baseline plasma value; PED = paediatric; DOSE = after dosing; METHOD = method paper, population unclear; MANUAL = manually entered; UNK = unclear; n/a = no value. Free-fraction correction is not applied (Table 4b).

Table 4. Concentration-realism filter (rule A: blood upper bound / activity ≥ 0.02) by metabolite class, total concentration.

Metabolite class	Not excluded (R ≥ 0.02)	Excluded (R < 0.02)	No concentration data	Pairs
Other	0	0	11	11
Endogenous/amino acids	43	17	3	63
Polyphenols	356	24	61	441
Bile acids	19	1	0	20
Fatty acids/SCFA	32	0	0	32
ALL	450	42	75	567

Note. 567 metabolite–target pairs (scheme A: the 107 E46 pairs whose SEA record was misattributed are removed from the original 674). “No concentration data” pairs are not excluded by this rule. The rule was applied to total, not unbound, concentration.

Table 4b. Sensitivity of the filter to an assumed unbound fraction (fu).

Subset	Pairs	Assumed fu	R·fu ≥ 0.02	R·fu ≥ 0.1	R·fu ≥ 1
All pairs with a concentration	492	1.0	450	345	194
All pairs with a concentration	492	0.1	294	194	124
All pairs with a concentration	492	0.01	180	124	76
Source class OK only	229	1.0	218	139	50
Source class OK only	229	0.1	106	50	28
Source class OK only	229	0.01	43	28	18
Source OK and 134-core target	138	1.0	131	78	26
Source OK and 134-core target	138	0.1	58	26	10
Source OK and 134-core target	138	0.01	20	10	4

Note. Counts of pairs with R·fu above the stated threshold (0.02 is the rule A cut-off). “Source class OK only” restricts to pairs whose blood value is an adult baseline plasma value. fu is a scenario, not a measurement: measured fu exists only for quercetin; for most flavonoids it is unknown.